Window Labels: Missing in Action

Thermal certification field labeling is simple, and inexpensive, when a knowledgeable representative from the window manufacturer is involved.

by Arlene Z. Stewart and Roland E. Temple

Over the last six months, we've encountered a number of thermal labeling problems with windows in our HERS ratings. These problems run the gamut from no thermal label at all to a partial label to an improper label. Unfortunately, the most serious problem we've encountered is invalid field labeling by a dealer or the manufacturer's representative who are unfamiliar with proper field labeling procedures. This is a serious problem because ultimately, the whole home certification could be considered compromised when these little-known procedures are circumvented. In reality, field labeling for thermal certification is relatively simple and inexpensive, provided the correct procedure is followed and a knowledgeable representative from the window manufacturer is involved.

CERTIFICATION PROCESS

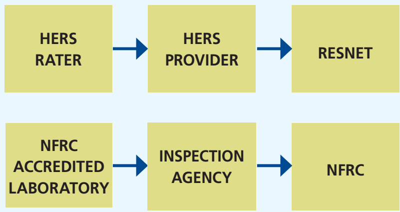

Figure 1. The certification process for NFRC is similar to the HERS process. The Lab checks the manufacturer, who is in turn checked by the Inspection Agency. NFRC oversees the whole process.

Window Labeling and the NFRC

In the late 1980s, the Federal Trade Commission (FTC) investigated dubious claims of highly efficient windows in Oregon and Washington State. In response, the National Fenestration Rating Council (NFRC) was formed in 1989 to provide independent verification of product performance so that the federal government would not have to develop standards and administer a certification and labeling program for the fenestration industry. The Energy Policy Act of 1992 (EPAct92) recognized the NFRC as the country's official fenestration energy performance certification and labeling organization, and it is cited in regulatory documents in 43 states.

NFRC Certification is a third-party process, not unlike HERS certification (see Figure 1). Fenestration manufacturers have their windows, doors, and skylights modeled, simulated, and tested by accredited, independent laboratories according to NFRC 100 and 200, when entering the program and every four years afterward, or when the product design is changed so as to alter performance. Next, NFRC-licensed Independent Certification and Inspection Agencies (IAs) validate and certify the ratings; they also visit each facility annually to ensure that the windows produced are labeled according to the test reports. Finally, the NFRC checks the IAs and the laboratories annually to ensure their compliance with the program.

It is the presence of the label that actually certifies the window. In fact, the NFRC Product Certification program (PCP) document contains specific language stating that a fenestration product shall not be NFRC-certified unless it has both the temporary and the permanent label affixed to it. Unfortunately, this knowledge is often restricted to the compliance engineer and does not filter down to sales staff or dealers. The energy rater should always check for both labels.

NFRC Temporary Label

The NFRC temporary label (see Figure 2) gives an overview of the product's energy performance at a glance. The energy performance values on the label represent the rating of the fenestration product (window, door or skylight) as whole systems (glazing and frame). The label contains mandatory energy performance information: U-factor, solar heat gain coefficient (SHGC) (see sidebar, "U-factor and SHGC"), and visible light transmittance. Product manufacturer identification information, including, but not limited to, manufacturer's name or identification code, frame type, product type, and glazing option, is also required. The label information is available on the NFRC's online Certified Product Directory at http://search.nfrc.org. This is a searchable database of NFRC-certified products.

NFRC Permanent Label

The NFRC permanent label (see Figure 3) is attached to the product to insure that it can be identified and verified as a certified fenestration product after the

NFRC TEMPORARY LABEL

Figure 2. Typical NFRC temporary label indicates the thermal performance of the fenestration product.

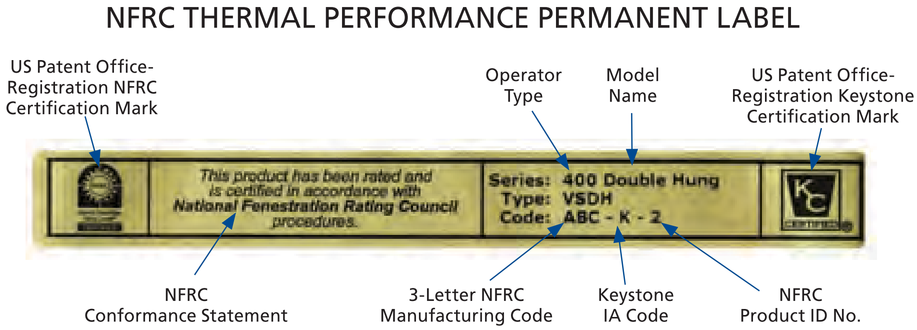

Figure 3. The typical permanent label indicates the NFRC certification mark, manufacturer, and IA information.

temporary label has been removed (after installation or certificate of occupancy or both). The permanent label should be attached to the product in such a way that it will remain attached: interior jam label, a tab or extension to an existing certification label, or a series of marks or etchings on the frame or glass. Any one, or combination, of these should contain all the required information. The permanent label requires the following mandatory information: the NFRC wordmark or the NFRC logo, the licensee's identification, and the product line identification. This information should make it possible to trace the window back to a particular Certification Authorization Report (CAR) in the Certified Product Directory, which contains performance characteristics of the line.

Labels and Performance

Today, computer modeling to estimate home energy performance is commonplace for both new construction and retrofits. HERS raters and weatherization evaluators can determine the best-performing window for a given application, and they can quantify the cost-benefit improvement to be realized by installing a window with the appropriate U-factor or SHGC. Window labels are essential for this purpose, because what you see is not necessarily what you get—most of the efficient technologies are invisible. Often the only visible sign of a window's energy performance is the number of panes or whether the glass is clear or tinted. However, windows that look identical might differ in their energy properties by as much as 75%!

Ideally, if efficient windows are expected on a job site, the construction manager should cross check the U-factor and SHGC with the actual order. If they don't match, the windows should be sent back and replaced with the right windows. Unlabeled windows should definitely not be accepted. Unfortunately, too often no one realizes that things aren't quite right until the window has been installed. Whole window replacement can have structural consequences, so a working knowledge of label troubleshooting is essential. The label will most often contain structural performance information, such as design pressure and impact resistant glazing. These structural performance values are vital to the location of the window in the building envelope.

Label Troubleshooting

The NFRC PCP spells out field-labeling discrepancies. Proper field labeling provides an alternative measure, and Section 9.1.4 lists three instances where these field discrepancies may occur:

1. A permanent or temporary label is missing or illegible.
2. The validity of the permanent or temporary label is in question.
3. The glass containing the label gets broken during transport or installation.

When a labeling discrepancy occurs, the PCP contains exact guidelines that the licensee and the IA must follow to correct the situation. These guidelines are contained in Section 9.1.4.1, "Corrective Actions On-Site" (http://nfrc.org/programdocs2.aspx).

U-factor and SHGC

The NFRC 100 procedure determines U-factor, the rate of heat loss through an assembly. While insulating value is indicated by the R-value (the inverse of the U-factor), R-value is not a valid measurement for windows. The FTC requires that window ratings use U-factors because each component has a different R-value, making windows complex assemblies. Thus a lower U-factor actually indicates more efficient performance.

The NFRC 200 procedure measures solar heat gain coefficient (SHGC), or how well a product blocks heat caused by sunlight. SHGC is the fraction of incident solar radiation admitted through a window (both directly transmitted and absorbed) and subsequently released inward. SHGC is expressed as a number between 0 and 1. The lower a product's SHGC, the less solar heat it transmits into the house.

Both of these measurements are critical to accurate HVAC sizing, occupant comfort, and long-term durability.

The most common discrepancy the energy rater will encounter will be a missing or illegible temporary label. When this occurs, you should look for the permanent label attached to the product. On operable products, such as hung and casement windows, the permanent label can be attached to the frame, or interior jam or to the sash when the product is in the open position. On fixed products, the label is attached to the frame in such a way that it is visible after installation. Once you locate the permanent label, the information on it should direct you to the licensee, the IA and the CAR, as described above.

If the permanent label is also missing, you must identify the manufacturer (licensee) of the fenestration product by contacting the builder or the dealer of record. Notified of the discrepancy, the licensee, in conjunction with the IA, must then verify the product on-site as the specified product.

Licensed Inspection Agencies

The NFRC-licensed Independent Certification and Inspection Agencies (IAs) listed below are responsible for reviewing and validating simulation and test reports of fenestration products and issuing Certification Authorization Reports (CARs) and conducting in-plant inspections.

American Architectural Manufacturers Association (AAMA)
Web site: www.aamanet.org
Tel: (847)303-5664
Primary contact: Dean Lewis
E-mail: dlewis@aamanet.org
Web: www. aamanet.org
Subcontractor: Associated Laboratories, Inc.
Primary contacts: John Vaughan or Brad Snoddy
Tel: (214)565-0593
E-mail: jvaughan@assoc-labs.com or brada@assoc-labs.com

Keystone Certifications, Incorporated
Tel: (717)932-8500
Primary contact: Marcia Falke
E-mail: mfalke@keystonecerts.com
Web: www.keytonecertts.com

National Accreditation and Management Institute (NAMI)
Tel: (757)594-8685
Primary contact: Sharon Durand
E-mail: namicertification@nami.hrcoxmail.com
Web: www.namicertification.com

Window and Door Manufacturer's Association (WDMA)
Tel: (847)299-5200 or (1-800)223-2301
Primary contact: John McFee
E-mail: jmcfee@wdma.com
Web: www.wdma.com

The licensee and the IA must follow the proper procedures for field labeling, as outlined in the PCP. Just bringing the right label to the site is not enough.

Sometimes the energy rater may find that the labels are in place, but the information is different. For example, if the product name or series number is listed on the temporary label, the same identification must be on the permanent label. The IA identified on the label should be contacted by e-mail immediately to take the necessary steps to correct the problem.

In the case of a broken pane, you will usually find the temporary and/or the permanent label in place. Notify the dealer of record of the broken pane, who should in turn contact the manufacturer for a labeled replacement pane or sash, according to the established PCP procedures.

Basically, if a manufacturer can trace the window order and composition back to the plant with certainty, the IA can give authorization to field label the product. To save time and cost, the builder, remodeler, HERS rater, or weatherization verifier can assist by providing a picture of the manufacturer's tracking label to the IA or the compliance engineer.

Remember that a fenestration product cannot be considered NFRC-certified unless both the temporary and the permanent labels are attached to the product. You must always look for both labels when a labeling discrepancy occurs.

Luckily, there are only four IA's in the NFRC program (see "Licensed Inspection Agencies"). So send each of them a quick e-mail explaining the situation, asking for the appropriate contact at the manufacturer in question. Within Appendix A of the PCP, the NFRC has the authority to levy fines for improper labeling—as much as $5,000 per product line, per product—so the compliance engineer at the manufacturer is often quite relieved to be notified directly. While the dealer should be notified that there is an issue, we do not recommend directly asking the dealer to provide a temporary label. The licensee and IA may ultimately ask the dealer to help correct the problem, but the dealer does not have the authority to certify a product that is missing a label. Dealers often have no knowledge of field-labeling procedures, and no contact with the compliance engineer, and they can become belligerent when the label they provide is not acceptable.

Proper Field Labeling

The NFRC procedures allow labeling discrepancies to be corrected on-site, provided the procedural criteria are met. The main concern is that the licensee's corrective actions identify the condition that created the discrepancy, and the licensee will provide the IA with the description of the corrective action. The corrective action must include some means of ensuring that the discrepancy will not recur. All documentation must be kept on file by the IA.

This documentation may include

- production control records that identify the product and glazing option;
- quality assurance records;
- a report of an on-site inspection by the licensee's employee or authorized agent to attest to the validity of the correction; and
- in the case of a broken pane, documentation affirming that the broken pane was replaced with correct glazing.

Another main stipulation of the corrective field-labeling procedure is that only products manufactured at a facility registered with the IA can bear the certification label, and only a licensee's employee or authorized agent shall be authorized to affix the label on-site. This stipulation ensures that the manufacturer and the IA have control of the certification labels, and that the proper labels are attached to the products. The NFRC and NFRC licensed IAs have the authority to suspend the licensee for noncompliant-labeled products.

In conclusion, whatever the labeling discrepancy, the energy rater or weatherization evaluator must receive confirmation from the licensee's IA that the NFRC field-labeling procedures have been followed, and that the proper U-factor and SHGC values have been verified and assigned for the product. Accepting a replacement label or component without IA authorization could compromise the whole home rating because the window is *not* a certified product.

Arlene Z. Stewart and Roland E. Temple represent the Efficient Windows Collaborative, a market transformation project funded by DOE. Stewart is a HERS rater, while Temple chairs the Certification Policy Committee for the National Fenestration Rating Council.